



Cambridge International AS & A Level

ECONOMICS

9708/32

Paper 3 Multiple Choice

February/March 2021

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **thirty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 30.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

Done

This document has **12** pages.

No one can be made better off without making someone else worse off

2

External EOS

- 1 Over time the average total cost curve of a firm is lowered. (Shift in ATC - dynamic)

Which form of efficiency does this represent? Word 'over time' is important.

A allocative (MC = price)

☒ B dynamic

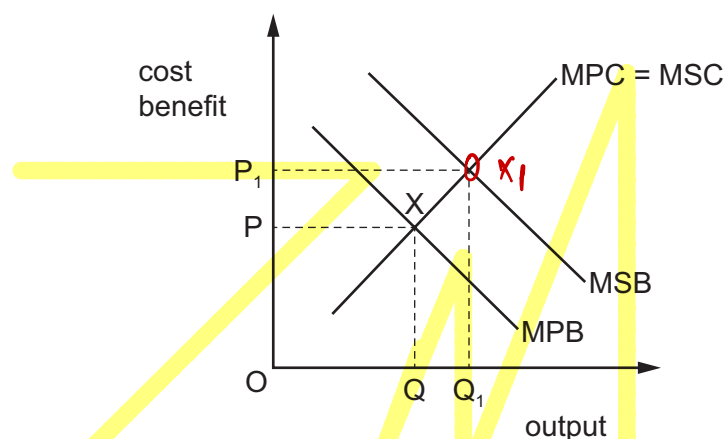
C Pareto

D productive

Dynamic efficiency refers to the ability of a firm to reduce its costs **over time** through innovations and improvements in production processes arising from re-investment of super-normal profits in research and development. A downward shift in the average total cost curve represents dynamic efficiency, as the firm is able to produce the same quantity of output at a lower cost.

ATC = min

- 2 The diagram shows the private and social marginal costs and benefits curves for the antibiotics market. The market equilibrium is at point X.



X: $MPB = MPC$
private equilibrium

X1: Socially optimum
 $MSB = MSC$

Why does market failure occur?

A There is overconsumption and overpricing.

B There is overproduction and under-pricing.

☒ C There is underconsumption and under-pricing.

D There is underproduction and overpricing.

(demand perspective) } Demerit goods
(supply perspective)
(under allocation of resources)
(supply perspective)

- 3 The table shows the results of a cost-benefit analysis of the construction of a new airport.

	US\$ million
private cost	400
private benefit	440
social cost	450
social benefit	480

$$SB - PB = EB$$

$$480 - 440 = 40$$

$$SC - PC = EC$$

$$450 - 400 = 50$$

What is the total value, in US\$ million, of all the externalities created by construction of the new airport?

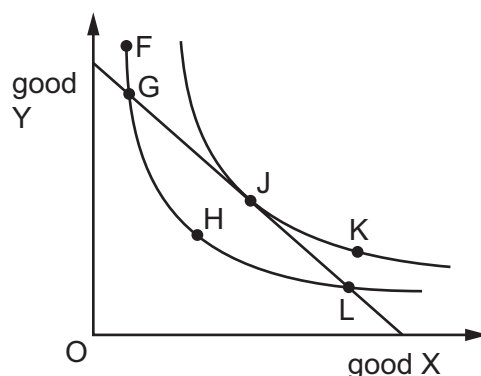
A 30

B 40

C 50

☒ D 90 (50 + 40 = 90)

- 4 A utility-maximising consumer has a fixed income and can choose between good X and good Y. The diagram shows her budget line and different consumption combinations (F, G, H, J, K and L) of products X and Y.



Which consumption combinations are currently available to her?

- A F, G, H and L
B F, G, J, and L
C G, J and L only
✓ D G, H, J and L

↳ On and inside budget line

↓

G, J, L

↓

H

- 5 Rational consumer Romesh spent 90 rupees buying herb X and spice Y. Both cost 10 rupees per gram.

MU_x / P_x	MU_x	grams purchased	total utility herb X	total utility spice Y	MU_y	MU_y / P_y
1.5	15	2	15	12	12	1.2
1.5	15	3	30	25	13	1.3
1.5	15	4	45	35	10	1
1	10	5	55	42	7	0.7

How much of each should he buy?

	herb X (grams)	spice Y (grams)
A	2	3
B	3	5
C	4	5
✓ D	5	4

$$\frac{MU_x}{P_x} = \frac{MU_y}{P_y}$$

↓
5 units

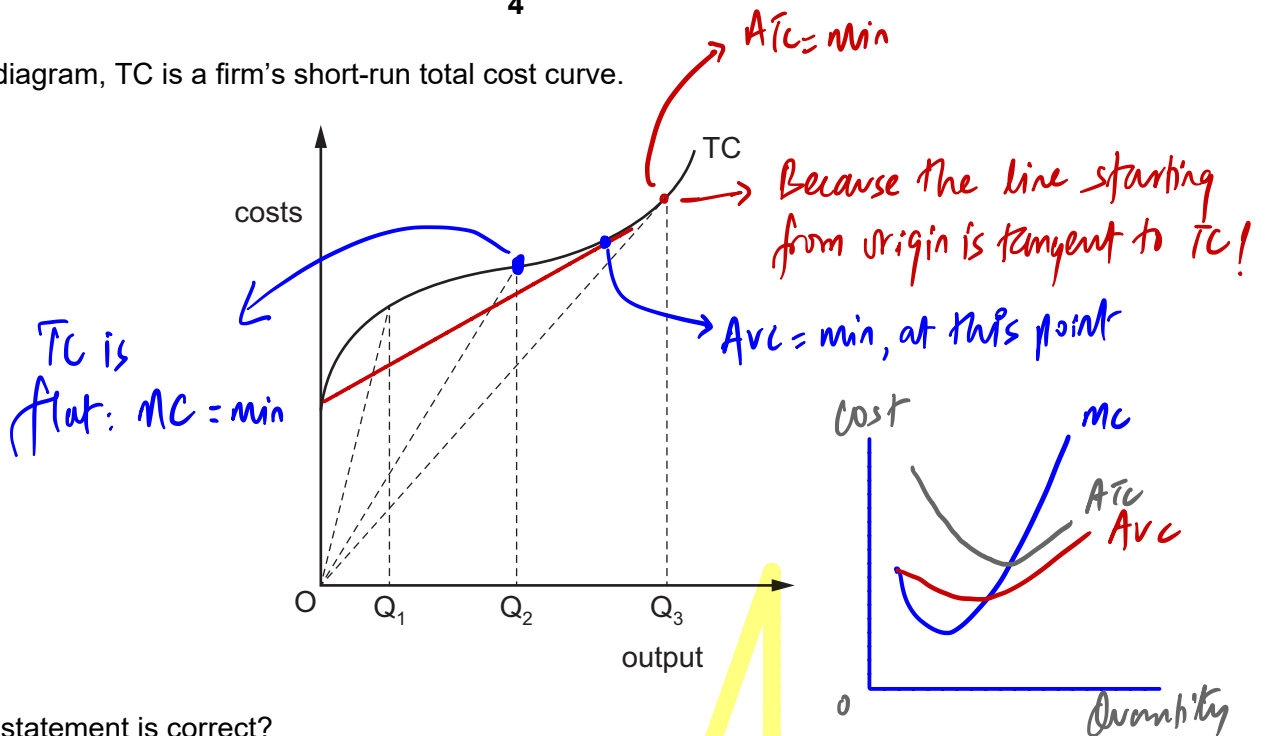
↓

4 units

[consumption bundle]

Equi-marginal principle

6 In the diagram, TC is a firm's short-run total cost curve.



Which statement is correct?

- ☒ A Average total cost is minimised at output QQ_3 . (correct)
- ☐ B Average variable cost is minimised at output QQ_2 . (Between Q_2 and Q_3)
- ☐ C Average variable cost is minimised at output QQ_3 . (same as option B)
- ☐ D Marginal cost is minimised at output QQ_1 . (at Q_2)

7 What is correct about cartels?

- ☐ A They are a form of trade union restrictive practice. Not related to trade union
- ☐ B They are a type of industrial merger. Merger is when 2 or more companies combine to form new entity
- ☒ C They are often subject to government investigation.
- ☐ D They establish a vertical link between firms.

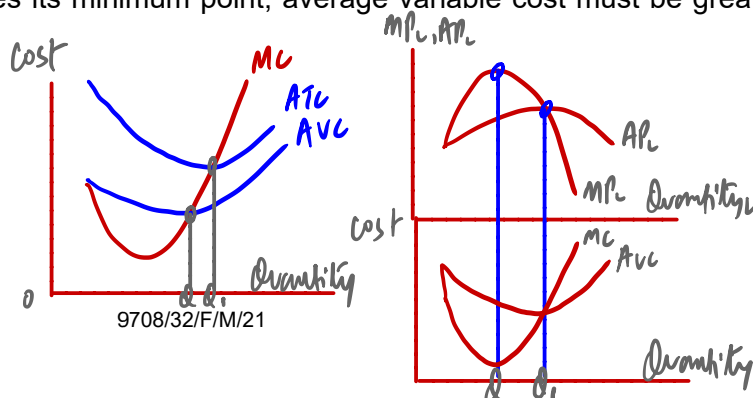
It establish horizontal link.

C - Cartels are agreements between companies to fix prices, restrict output, or engage in other anticompetitive behaviors. They are illegal in many countries and are often subject to investigation and penalties by government agencies responsible for enforcing antitrust or competition laws.

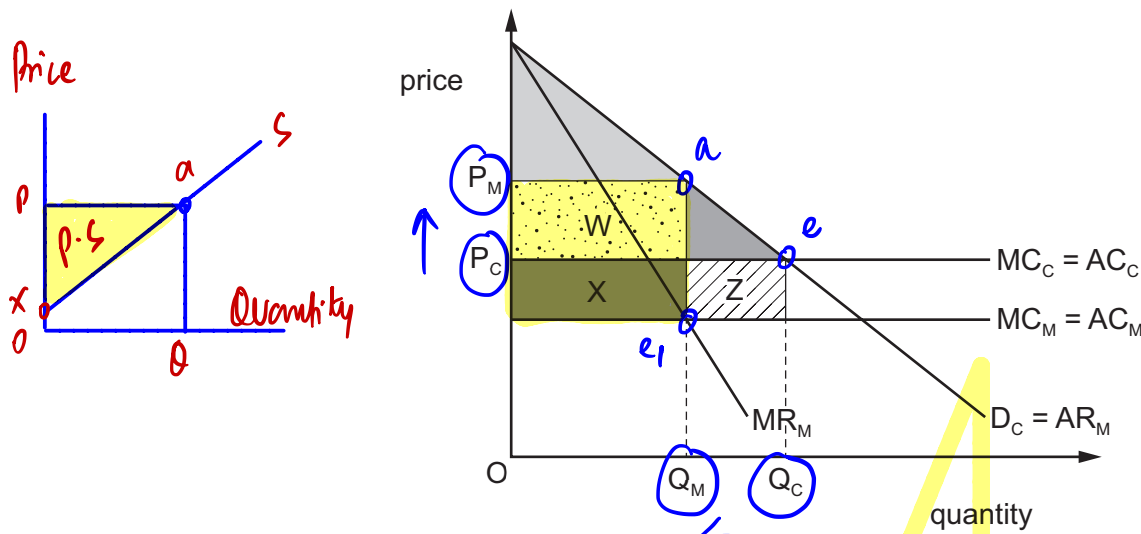
8 When output increases in the short run, which statement is correct?

- ☐ A Average fixed cost first falls and then, beyond some point, rises. No, AFC falls only, TFC/Q , $Q \uparrow AFC \downarrow$.
- ☐ B Average variable cost increases as soon as the law of diminishing marginal returns begins to operate.
- ☒ C The minimum point of the average total cost curve occurs at a greater level of output than the minimum point of the average variable cost curve. Correct. ATC min comes after AVC min.
- ☐ D When marginal cost reaches its minimum point, average variable cost must be greater than average fixed cost.

Law of diminishing returns operate when $MP_L \downarrow$.
 $AVC \uparrow$ when $AP_L \downarrow$,
 See mirror picture!



- 9 The diagram shows a competitive industry facing constant marginal and average costs. In equilibrium its output is OQ_C and its price is OP_C . A series of horizontal mergers generate economies of scale and create a monopoly in the industry. The new equilibrium output is OQ_M and price is OP_M .



Which area represents the producer surplus after these mergers?

- A W ☒ B W + X C W + X - Z D X

- 10 A firm that controlled over 50% of the market successfully merges with its main competitor to form one large company.

What is the most likely reason why, from the firm's point of view, the merger will fail?

- A a fall in consumer surplus (worsens consumers, benefits producer)
☒ B cost of funding the merger (only reason for failure) (cash flow problems) (liquidity issues)
 C economies of scale (ATC ↓)
 D market dominance (Monopoly power)

Benefits the firm

11 The diagram shows a monopolist's cost and revenue curves.

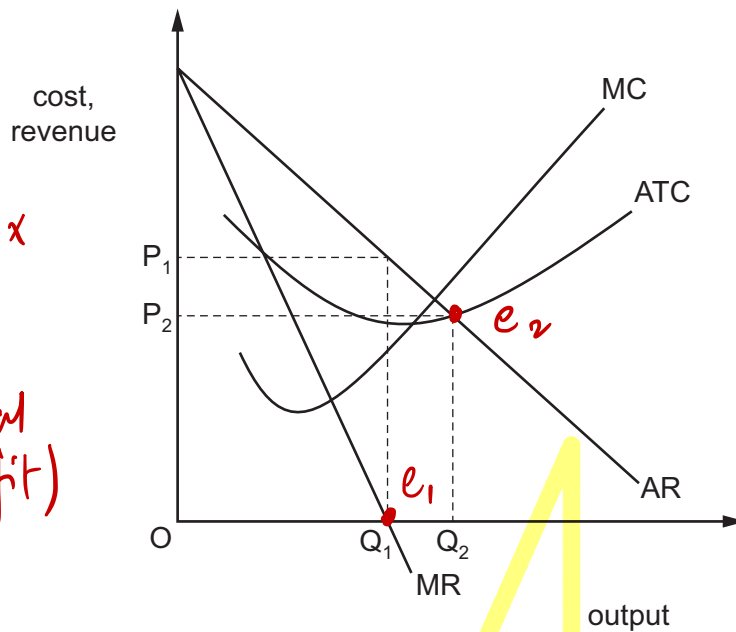
$$Q_1 : e_1$$

$$MR = 0$$

Sales revenue max

$$Q_2 : e_2$$

ATC = AR (Normal profit)
Sales max



The monopolist changes its price from P_1 to P_2 and its output from Q_1 to Q_2 .

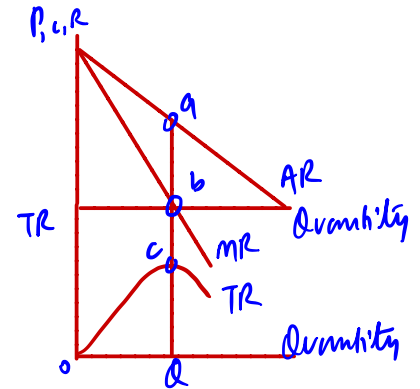
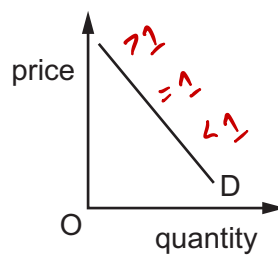
Which change in objective is indicated by the move from P_1 to P_2 ?

- MC = MR
- A** profit maximisation to sales revenue maximisation
- B** profit maximisation to sales maximisation subject to earning a normal profit
- MC = MR
- C** sales revenue maximisation to profit maximisation
- MR = 0 AC = AR
- D** sales revenue maximisation to sales maximisation subject to earning a normal profit

12 The diagram shows the demand curve for a firm's product.

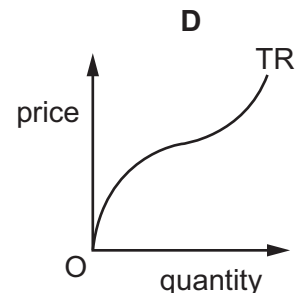
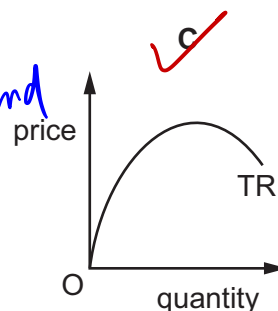
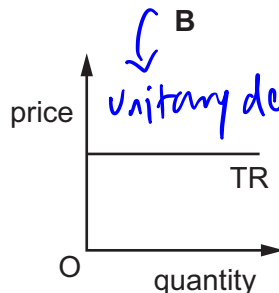
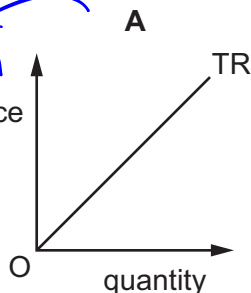
* Along the demand curve, PED is different at each price level.

* Intro to imperfect competition class.



Which diagram depicts the shape of the firm's corresponding total revenue (TR) curve?

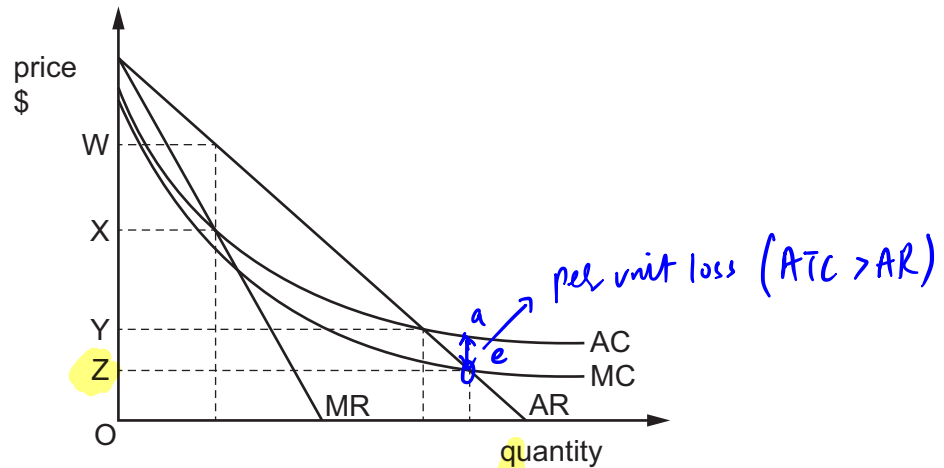
perfect competition



* Important, y-axis should be TR!

$$MC = AR$$

13 A government is attempting to achieve allocative efficiency in a nationalised industry.



What will the government do?

- A It will set price at W and make sub normal profit.
- B It will set price at X and the firm will make supernormal profits.
- C It will set price at Y and make normal profits.
- ✓ D It will set price at Z and make a loss.

None meets the condition of allocative efficiency, that is $MC = AR(P)$

14 The diagram shows a sign beside a busy road.



What is this an example of?

- A free good (unlimited in supply, no opportunity cost)
- B government regulation (laws/limit/ban)
- C merit good (underconsumption, underproduction, underallocation)
- ✓ D nudge theory

D - Nudge theory is a concept from **behavioral economics** that suggests **influencing people's behavior and decisions without mandating or restricting their choices**. The example of the notice on the motorway near Cambridge is an illustration of **nudge theory**.

-> Nudge theory involves **providing information** and by **highlighting that clearing litter puts road-workers' lives at risk**, the notice aims to **increase individuals' awareness of the potential negative consequences of their actions** that will encourage them to **make choices that are in their best interest** or align with desired societal outcomes. In this case, the notice about litter being cleared is likely intended to **remind drivers not to litter**, thus **encouraging a change in behavior** to keep the environment clean.

real income ↑

8

- 15 Recent research has shown that consumption of some foods might be harmful. A government committee considers how to intervene in the markets.

Which intervention is most likely to reduce consumption of these foods without reducing income equality?

- A a minimum price for the harmful foods, Minimum price, price ↑, consume ↓, real income ↓
- ✓ B a subsidy for substitute foods, cheaper alternative, harmful food consumption ↓
- C an ad valorem tax on the harmful foods } Indirect tax, P ↑, consume ↓
- D an indirect tax on the harmful foods } real income ↓

Doesn't represent poverty trap

- 16 In which circumstances is a poverty trap least likely to operate?

- ✓ A A female worker is paid less than a male worker for doing the same job. Gender discrimination
- B A worker is paid the minimum wage and refuses a higher paid job because net income is lower after tax. Net income is lower
- C A worker refuses higher paid work as it means more expensive travel and childcare costs. More expenses
- D An unemployed worker refuses a job because earnings are less than unemployment benefit.
↳ Earning will be less after accepting job.

- 17 What would be most likely to cause the demand for labour in an industry to be inelastic?

- ✓ A Labour costs are a small percentage of total cost. Rise in wage will not raise TC by greater %, so quantity demanded will contract by smaller %.
- B The demand for the final product has a price elasticity of demand that is greater than one.
- C The workforce belongs to a strong trade union.
- D There is a large pool of readily available labour. Surplus of labours, elastic demand

Doesn't make D_L inelastic

{ Demand for product (>1)
↳ Demand for worker (>1)

- 18 Which influence is least likely to promote sustainable economic growth in a developing economy?

- A high saving rates, save ↑, investment ↑, growth ↑
- B involvement in free trade, cheaper imports, raw material cheap, cost ↓, AS ↑
- C protected property rights → Specialization, less wastage
- D rapid population growth

↳ dependency ↑

C) Protected property rights are essential for promoting sustainable economic growth in a developing economy. Secure property rights can incentivize landowners and businesses to invest in environmental protection and sustainable practices, as they have a long-term stake in the property and want to maintain its value.

→ A poverty trap refers to a situation where an individual or household is discouraged from working or earning more income because the financial benefits of doing so are reduced or eliminated by various factors, such as taxes, loss of benefits, or increased expenses.

- 19 Governments are interested in providing statistics on the standard of living. They measure the value of material goods and services.

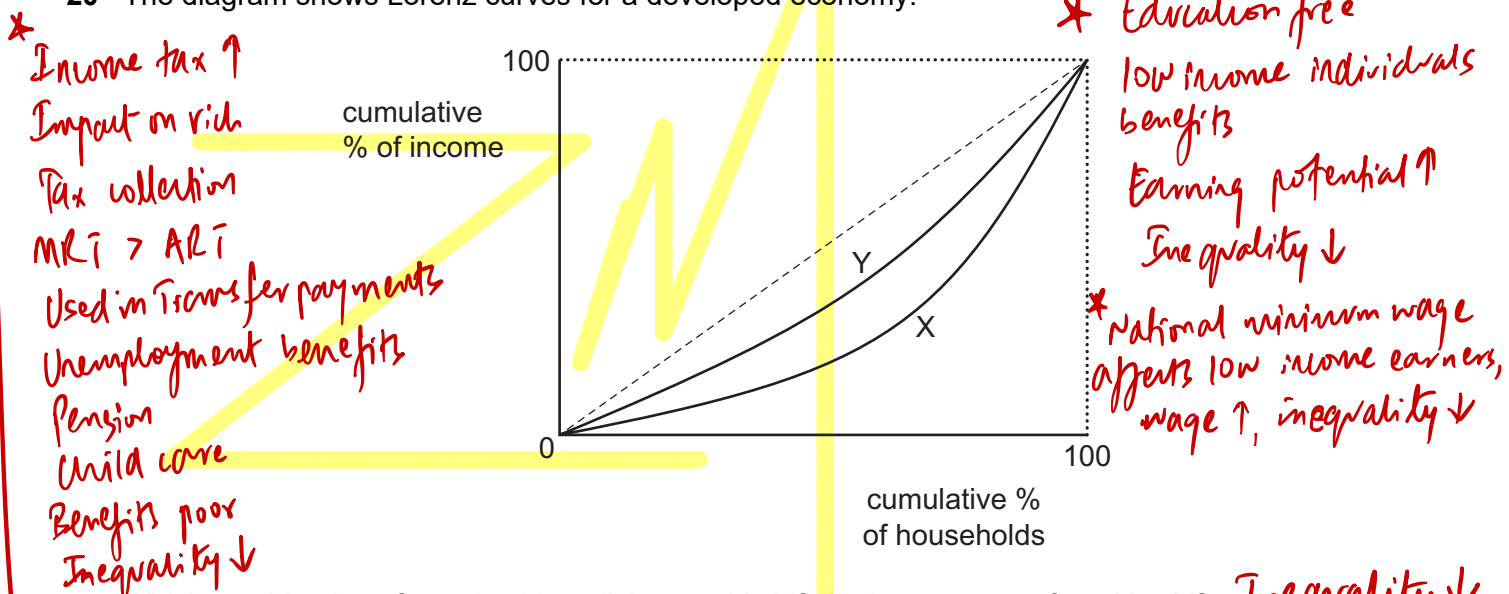
Non-material characteristics that affect the standard of living cannot easily be measured.

What is an example of a non-material characteristic?

- A the failure to include unpaid work in the home *Non-market activity*
 B the freedom to make informed economic choices
 C the proportion of GNP used for weapons of war *material aspect of government spending*
 D the removal of taxes from all forms of consumer expenditure

part of fiscal policy, affects cost of living, not a non-material characteristics

- 20 The diagram shows Lorenz curves for a developed economy.



Which combination of supply-side policies would shift the Lorenz curve from X to Y? *Inequality ↓*

	top rate of income tax	education	national minimum wage
☒ A	increase ✓	free ✓	increase ✓
B	increase ✓	students pay	abolish
C	reduce	free ✓	abolish
D	reduce	students pay	increase ✓

Non-material characteristics refer to aspects of well-being that go beyond material goods and services, and are difficult to quantify or measure. Examples include:

- Freedom and autonomy
- Social connections and relationships
- Environmental quality
- Personal fulfillment and happiness

Structural (fall in demand of particular industry)

- 21 Two members of the Hernandez family have recently become unemployed. Mariana has lost her job at the local factory, which has closed because of competition from cheaper imports. Alfonso has lost his job as a shop assistant because the sales have dropped during a recession.

How would an economist classify these job losses?

	Mariana	Alfonso
A	frictional unemployment	seasonal unemployment
B	regional unemployment	frictional unemployment
C	structural unemployment	cyclical unemployment
D	technological unemployment	structural unemployment

Over-specialization

in between jobs

e.g. outdated skill

→ AD↓ (cyclical demand deficient)
→ particular time of a year, e.g. tourism

- 22 Which combination of fiscal, monetary and supply-side policies is most likely to reduce unemployment?

	fiscal policy	monetary policy	supply-side policy
A	increasing taxation AD↓	increasing interest rate AD↓	increasing flexibility in the labour market ✓
B	increasing government expenditure ✓	decreasing the money supply AD↓	increasing education and training
C	decreasing taxation ✓	decreasing interest rate ✓	increasing flexibility in the labour market ✓
D	decreasing government expenditure AD↓	increasing the money supply ✓	increasing education and training

Reducing trade union power, minimum wage reduced or removed, wage ↓, $Q_d \uparrow$, employment ↑

Fiscal:

Tax ↓, $C \uparrow$, $I \uparrow$, $AD \uparrow$

+ $G \uparrow$, $AD \uparrow$:
Expansionary.

Monetary: $M \uparrow$, $r \downarrow$, credit regulation ↓: Expansionary

- 23 Which policy combination will reduce the level of national income by the greatest amount?

A decrease direct taxation and increase interest rates $C \downarrow$, No effect

B decrease direct taxation and increase quantitative easing Expansion

C increase direct taxation and increase interest rates Contractionary fiscal and monetary: $AD \downarrow$

D increase direct taxation and increase quantitative easing, No effect.

$C \downarrow$

$C \uparrow$

- 24 In a closed economy, the marginal propensity to save is 0.1 and the marginal propensity to pay taxes is also 0.1. These values are constant and do not vary with the level of income.

What will be the increase in national income if there is an injection of \$100 million into the circular flow?

A \$1000 million

✓ B \$500 million

C \$100 million

D \$80 million

$$MPS = 0.1$$

$$MPT = 0.1$$

$$\frac{1}{mps + mpr} \times AE$$

$$S \times 100$$

$$\$ 500 m$$

$$\frac{1}{0.1 + 0.1} \times AE$$

$$\frac{1}{0.2} \times 100$$

Speculative: hold more cash when interest rate are low and vice versa.

11 Money demand

25 Which combination is consistent with liquidity preference theory?

	type of demand for money	determined by interest rates	determined by changes in income
A	speculative ✓	yes ✓	no ✓
B	speculative ✓	no	yes
C	transactions	no	no
D	transactions	yes	no

associated with interest rate and changes in income

26 What is assumed by Keynesians?

- A An increase in supply always creates an equal increase in demand. No, Keynes argue AD is not sufficient.
- B Government intervention is required to manage the economy. ✓
- C Prices always adjust so that markets clear. Monetarist
- D The economy always has full employment. Monetarist, related to LRPC, NAIRU

Correct, role of government to stabilize economy.

27 Which row correctly identifies injections into a country's circular flow of income?

	private sector (S > I)	public sector (T > G)	trade sector (X > M)
A	no	no	no
B	no	no	yes
C	yes	no	no
D	yes	yes	yes

no no yes
Saving, Tax, imports: W
Investment, Government, Exports: Injections (J)

28 Which policy by developed economies would be most beneficial for developing economies?

- A a ban on resource exploration by mining companies in developing economies worsen (Ban)
- B restrictions on investment by global firms in developing economies worsens (restriction)
- C the removal of restrictions on emigration of skilled labour from developing economies
- D the withdrawal of agricultural subsidies in developed economies

Developed countries moving towards free trade, Even playing field for developing countries in international market.

skilled workers leaving developing economies, worsens

29 The natural rate of unemployment in an economy is 5%.

What will happen if a government persists in trying to achieve a target rate of unemployment of 3% by expansionary monetary policy? $AD \uparrow > AS$: price $\uparrow$

- ☒ A an accelerating rate of inflation $\Rightarrow$ augmented Phillips curve
- B a diminishing rate of inflation
- C a high but constant rate of inflation
- D a negative rate of inflation (Deflation)

Expansionary

30 What is a reflationary fiscal measure?

- ☒ A increasing government spending $G \uparrow, AD \uparrow$
- B increasing taxes $\rightarrow C \downarrow, AD \downarrow$
- C increasing the money supply } Monetary
- D reducing interest rates

1. **Government tries to reduce unemployment below NAIRU:** The government uses expansionary monetary policy to stimulate the economy and reduce unemployment below the natural rate (NAIRU).
2. **Reduction in unemployment:** Unemployment falls as the economy grows and more people find jobs.
3. **Upward movement along SRPC (higher inflation):** As unemployment falls, the economy moves upward along the Short-Run Phillips Curve (SRPC), and there is a rise in demand pull inflation.

Higher inflation results from the increased demand for labor and resources. This shows the trade-off between unemployment and inflation.

4. **Short-run Philips curve shifts towards the right:** When there is a mismatch between wage increases and increases in productivity, there will be an increase in costs of production, as a result of which, Aggregate supply will fall, leading to a rightward shift in the short run Philips curve. This indicates an increase in the rate of inflation and also unemployment as the economy moves back to the natural rate of unemployment.
5. **Long-run adjustment: economy returns to NAIRU, inflation stabilizes at a higher level:** Eventually, the economy adjusts to the demand-side shocks, and the labor market returns to equilibrium at the natural rate of unemployment.

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